

Centre for Distance and Online Education

MCA

Semester - III

Cyber Security (CS)



Prepared by

Dipak Dhariyaparmar

Enrollment no: - 24227521010008

Faculty Name

Jay Parmar



Theory Question

Q-1: - What are the primary types of cybersecurity threats, and what strategies can organizations implement to mitigate these risks?

Ans: -

=> Today almost everything is connected to the internet. We use online banking, shopping, social media, and even office work is done online. Because of this, cyber threats have also increased.

Cybersecurity threats are basically problems or attacks that try to steal, damage, or misuse data and systems. To stay safe, it is important to know the main types of threats and how to stop them.

Main Types of Cybersecurity Threats

1. **Malware:** - Malware means “malicious software.” These are harmful programs like viruses, worms, and ransomware. They can damage your computer, lock your files, or steal your information. For example, in ransomware, attackers lock all your data and ask for money to unlock it.
2. **Phishing:** - In phishing, attackers send fake emails, messages, or links that look real. Their goal is to trick you into giving your password, bank details, or personal information. For example, you may get an email that looks like it’s from your bank asking you to “verify your account.”
3. **Denial of Service (DoS/DDoS):** - In this type of attack, hackers send too much traffic to a website or server so that it becomes slow or even crashes. Companies lose business and trust when their websites don’t work properly.
4. **Insider Threats:** - Not all attacks come from outside hackers. Sometimes employees or staff members misuse their access to steal data or harm the system. Even a small mistake, like sharing the wrong file, can also cause damage.
5. **Password Theft:** - Many people still use weak or the same password for many accounts. Hackers can easily guess or steal them using special tools. Once they get the password, they can access important data and accounts.

Ways to Reduce Cybersecurity Risks

1. **Firewalls and Antivirus:** - Firewalls block unwanted access, and antivirus helps to detect and remove harmful software. This is the first step in protecting any computer system.
2. **Software Updates:** - Hackers often attack systems that are not updated. That’s why it is important to keep software and applications updated with the latest security patches.
3. **Strong Passwords and Two-Factor Login:** - Passwords should be long and strong, not simple ones like “12345.” Adding two-factor authentication (like OTP on your phone) makes it harder for hackers to break in.



4. **Employee Training:** - Many attacks happen because people click on fake links or emails. If employees are trained to identify such threats, they can avoid falling into traps.
5. **Regular Backups:** - Keeping backup copies of data is very helpful. If data is lost or locked by ransomware, it can be restored from the backup.
6. **Monitoring and Quick Action:** - Companies should keep an eye on their networks all the time. If anything suspicious happens, they should act quickly to reduce the damage.
7. **Limit Access:** - Not everyone should have access to all data. Giving access only to those who really need it reduces the chances of misuse.

Q-2: - How do advanced persistent threats (APTs) differ from traditional cyberattacks, and what strategies can organizations employ to detect and defend against APTs?

Ans: -

=> Cyberattacks are a major issue in today's digital world, but not every attack is the same. A **traditional cyberattack** usually happens quickly and is designed to cause immediate damage or steal information in a short time. For example, a phishing mail may trick someone into sharing their password, or malware might instantly corrupt or delete important files.

In comparison, an **Advanced Persistent Threat (APT)** is a completely different type of attack. It is long-term and very well-planned. Instead of openly attacking, the hackers secretly enter a system and stay hidden for weeks, months, or even years. During this time, they quietly steal sensitive data, monitor activities, or take control of parts of the system. APTs are usually carried out by highly skilled attackers, often linked with professional groups or even government agencies. Their aim is not quick profit, but long-term access and spying.

Main Differences:

- **Time:** Traditional attacks are short and quick, while APTs last for a long time.
- **Goal:** Traditional attacks look for fast results (like stealing money), while APTs want long-term access and secret data.
- **Visibility:** Traditional attacks are easier to notice, but APTs are very silent and hidden.
- **Target:** Traditional attacks usually target many random users, but APTs focus on specific organizations, industries, or governments.

How to Detect and Defend Against APTs

Because APTs are advanced and hidden, organizations need stronger methods to protect themselves. Some important strategies are:

1. **Constant Monitoring:** Companies should keep a close watch on their networks and systems. Unusual behavior, like strange logins or data movement, could mean an APT is present.



2. **Multiple Layers of Security:** Instead of only using antivirus, companies should also use firewalls, intrusion detection systems, and endpoint protection together.
3. **Threat Intelligence:** Organizations can use global reports and tools that provide information about new hacking techniques and APT groups.
4. **Employee Training:** Many APTs start with targeted phishing emails (called spear-phishing). If employees are trained to recognize suspicious emails and links, the chances of success for attackers go down.
5. **Access Control:** Not every employee should have access to everything. Limiting access to only what is required reduces risk.
6. **Regular Updates:** Hackers often take advantage of weak or outdated software. Updating systems regularly and fixing security issues helps close those doors.

Incident Response Plan: Even with good protection, attacks can still happen. Having a clear plan for what to do if an APT is found helps reduce damage and recover faster.

Q-3: - In the context of recent high-profile ransomware attacks, what measures can organizations take to prevent such incidents, and how should they respond if an attack occurs?

Ans: -

=> Ransomware attacks have become one of the biggest cybersecurity problems in recent years. In such attacks, hackers lock or encrypt an organization's data and demand money (ransom) to unlock it. Big companies, hospitals, and even government agencies have faced these attacks. To stay safe, organizations need both preventive steps and a proper response plan.

Measures to Prevent Ransomware Attacks:

1. **Regular Backups:** Keep multiple copies of important data in secure offline storage, so even if files are locked, they can be restored.
2. **Strong Security Tools:** Use updated antivirus, firewalls, and intrusion detection systems to block malicious activities.
3. **Software Updates:** Apply patches and updates regularly to close security gaps that hackers might exploit.
4. **Employee Awareness:** Train staff not to click on unknown links or open suspicious email attachments, since ransomware often spreads through phishing.
5. **Access Control:** Limit admin rights and ensure employees only access what they truly need.
6. **Email Filtering:** Use spam filters and email security tools to block phishing emails carrying ransomware.



How to Respond if an Attack Occurs:

1. **Isolate the Infected System:** Disconnect affected devices from the network to stop the ransomware from spreading.
2. **Do Not Pay Immediately:** Paying ransom does not guarantee data recovery and may encourage more attacks.
3. **Inform Security Teams:** Report the incident to cybersecurity teams and, if required, law enforcement.
4. **Restore from Backups:** If clean backups are available, restore systems instead of paying attackers.
5. **Incident Analysis:** After recovery, investigate how the attack happened and strengthen security to prevent future incidents.

Practical Question

Q-1: - Explain the steps of Understanding and monitoring, tracing received mail

Ans: -

=> When we receive an email, especially if it looks suspicious, we need to carefully **understand, monitor, and trace it** to check whether it is genuine or malicious.

The steps are:

1. Check the Sender's Email Address

- Look closely at the sender's email ID.
- Many phishing mails look like they are from trusted sources, but small spelling mistakes or unusual domains (like abc@paypa1.com instead of abc@paypal.com) can reveal a fake mail.

2. Examine the Subject and Content

- Notice if the subject line is urgent, threatening, or too attractive (e.g., "You won a prize").
- Check for poor grammar, spelling errors, or unusual requests like sharing passwords or bank details.

3. Inspect Email Headers



- In Gmail/Outlook, you can open the “Show Original” or “View Source” option to see the full header.
- Here you can trace details like Received From, IP addresses, mail servers used, and timestamps to know the real origin of the mail.

4. Check Links Before Clicking

- Hover the mouse over any link without clicking.
- If the link address doesn't match the official website (e.g., shows a strange URL), it's a sign of phishing.

5. Scan Attachments Safely

- Do not open attachments directly.
- First, scan them with an antivirus or use an online scanner (like VirusTotal). Many malicious mails carry malware inside attachments.

6. Trace the Mail Path (Advanced Check)

- By analyzing the Received lines in the header, we can trace the IP address of the sending server.
- This helps to find out the geographical location of the sender and whether it matches the claimed identity.

7. Report or Block

- If confirmed as malicious, mark the email as spam/phishing in your mail service.
- Inform the IT/security team so they can block such mails and protect other users.

Q-2: - Write the steps for Evaluating network scanning using Nmap Tool.

Ans: -

=> Steps for Evaluating Network Scanning using Nmap Tool

1. Open the terminal or command prompt where Nmap is available. If not installed, use the command:
`sudo apt-get install nmap`
2. To scan a single host, type:
`nmap <IP address>`



3. To scan multiple hosts within a range, use:
`nmap 192.168.1.1-50`
4. To scan an entire subnet, use:
`nmap 192.168.1.0/24`
5. To identify open ports and running services, use:
`nmap -sV <IP address>`
6. To detect the operating system of a target, use:
`nmap -O <IP address>`
7. For a detailed scan with combined options, use:
`nmap -sS -sV -O <IP address>`
8. Analyze the output displayed, which shows open ports, services, and system details.

Q-3: - Write the steps for Evaluating network sniffing with Wireshark.

Ans: -

=> **Steps for Evaluating Network Sniffing using Wireshark**

1. **Install Wireshark**
 - `sudo apt-get install wireshark`
2. **Launch Wireshark**
 - Open Wireshark from the applications menu or terminal.
3. **Select a Network Interface**
 - Choose the active network interface (e.g., Ethernet, Wi-Fi) but want to capture traffic from.
4. **Start Capturing Packets**
 - Click on the **Start Capturing Packets** button (blue shark fin icon).
 - Wireshark will begin capturing live network traffic.
5. **Apply Capture Filters (Optional)**
 - Use filters (e.g., `tcp, udp, http, ip.addr == 192.168.1.10`) to focus only on specific traffic.
6. **Monitor and Analyze Packets**



- View captured packets in real-time.
- Click on any packet to view its details, including source, destination, protocol, and payload.

7. Stop Capturing

- Click the **Stop** button (red square icon) when you have collected enough data.

8. Apply Display Filters

- Use display filters (e.g., http, dns, ftp) to analyze only the relevant traffic after capture.

9. Save the Capture File

- Save the captured traffic for future analysis:
 - Go to **File > Save As** and store in .pcap format.

10. Review and Evaluate Data

- Analyze packet details to identify suspicious activity, monitor network performance, or troubleshoot issues.